

Exempt, Evade, Enforce:
Unpacking Property Tax Dynamics
in the Short Term Rental Market



Darla Moore School of Business

Kara Jones

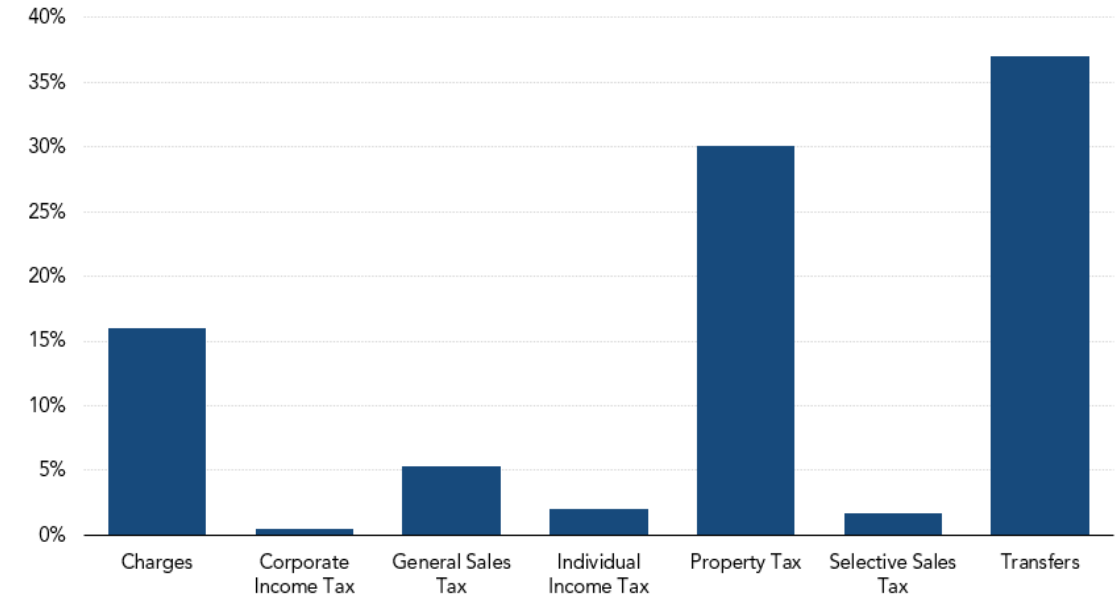
Motivation: The Hidden Backbone of Local Finance

- The property tax is the primary funding source of US counties
- Vulnerable to evasion due to self-reported property use
- STR hosts can strategically manage rental activity to retain owner-occupied exemptions
- Third-party reporting has the potential to play a decisive role in host behavior and tax revenue

FIGURE 3

Sources of Local General Revenue

Share of total local general revenues, by source, 2021



Source: US Census Bureau Annual Survey of State and Local Government Finances, 1977-2021 (compiled by the Urban Institute via State and Local Finance Data: Exploring the Census of Governments; accessed 14-Jul-2023 10:05), <https://state-local-finance-data.taxpolicycenter.org>.



Darla Moore
School of Business

Motivation: A home becomes a property tax loophole

- Large tax breaks for owner-occupied homes
 - Homestead Exemption (HE)
 - Legal Residence Exemption (LRE)
- No annual verification
- Rental Day Limits tied to exemptions
- Potential Response to Taxes:
 - **Comply**
 - **Avoid:** rent just below limits
 - **Evade:** claim exemptions illegally



Research Questions

1. How do short-term rental (STR) hosts avoid property taxes by manipulating the number of nights they rent?
2. Does Airbnb's third-party tax reporting increase compliance?

Identification Strategy: Bunching at Rental Day Limits

- Rental day limits for the homestead exemption
- FL: ≤ 30 days
- Exceed limit $\rightarrow$ increase tax bill
- Introduces a tax notch
- **Method:** Bunching Estimator
- **Data:** AirDNA data, FL, 2015-2024
- **Results Preview**



Contribution to Tax Evasion and Sharing Economy Literatures

1. Tax Compliance Across Regimes

- Compliance varies based on tax structure and enforcement point (*Marion 2008; Bieber 2021*)
- **STR response to a property tax thresholds, extending compliance beyond income and sales taxes**

2. Sharing Economy & STR Externalities

- STRs reduces hotel revenue and housing affordability while providing positive externalities (*Barron et al. 2021; Gurran & Phibbs 2017; Zervas et al. 2017; Franco et al. 2021*)
- **Link property tax policy to STR behavior and show how compliance can recoup public costs**

3. STR Regulation and Tax Policy

- STR regulations have mixed impact on STR listings and prices (*Garz 2023; Bei et al. 2023; Li et al. 2022*)
- **Use a natural rental day cutoff to understand STR behavior and inform enforceable, targeted local tax policy**

Motivation

Background

Empirical Strategy

Data

Results

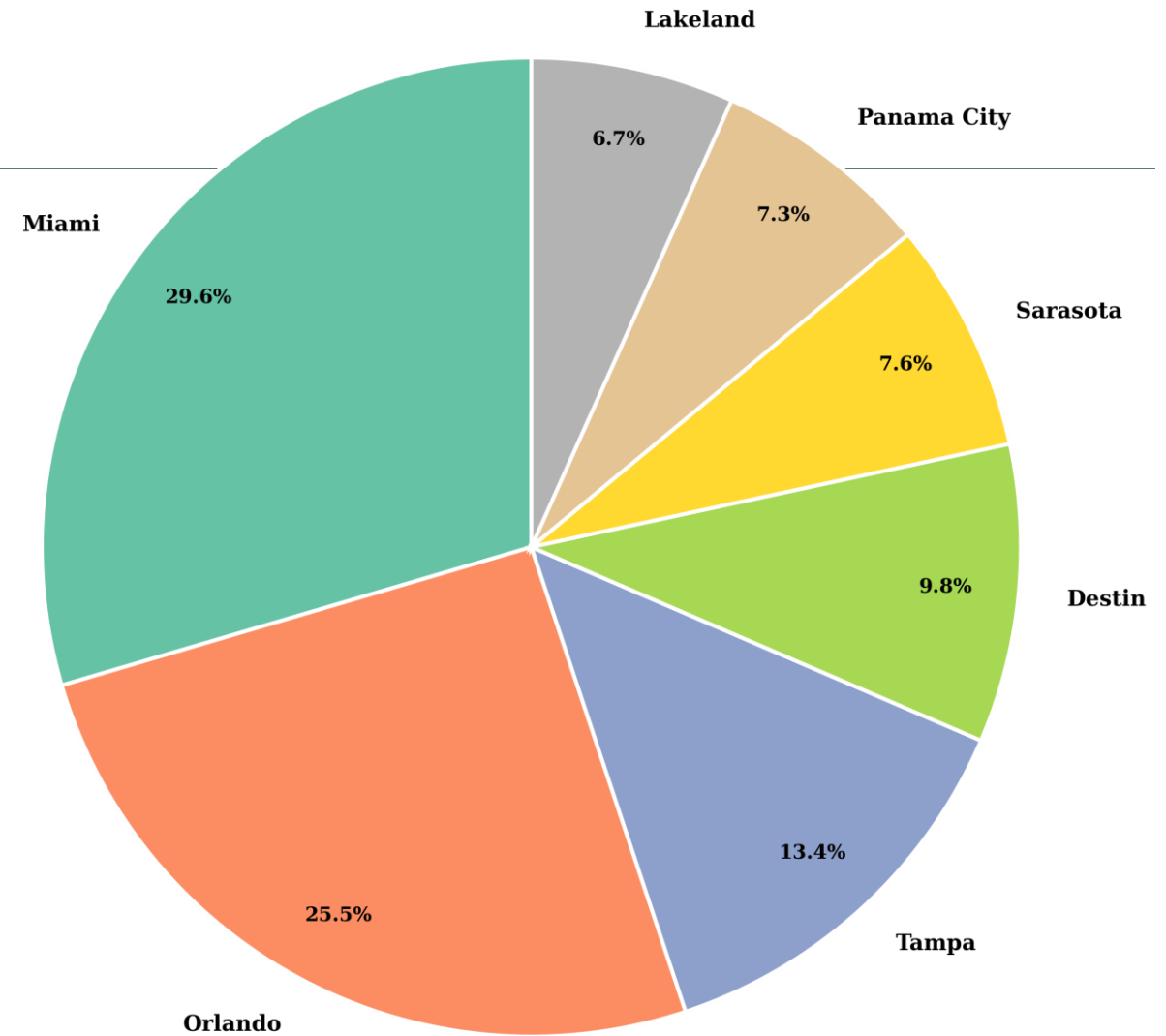
Conclusion



**Darla Moore
School of Business**

STR Market in Florida

- Over \$12 billion in revenue in 2024
- Approx. 10.5 million reservations in 2024
- STRs listed for an average of 3.09 years
- Most STR hosts own 21+ units (39%)
 - 1 unit: 37%
 - 2-5 units: 15%



Florida Property Tax Liability

- Home value: H
- Assessed Value: A
- Millage rate: M

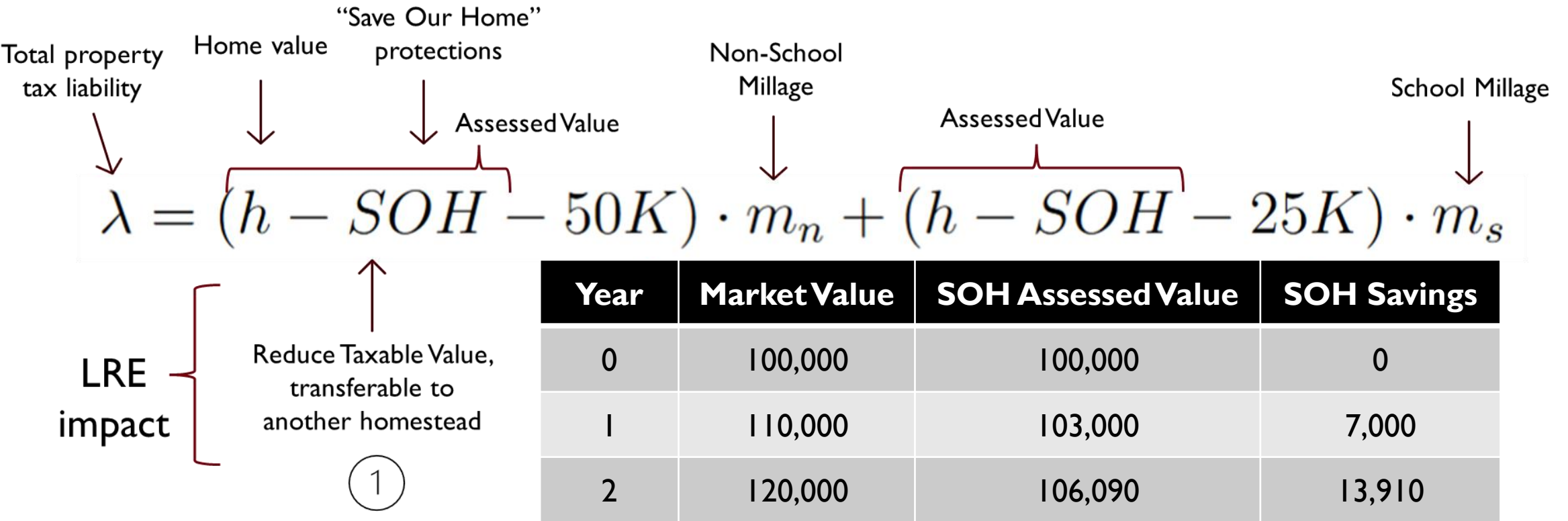
- $\text{Tax} = A * M$
- Where $A = f(H)$

Florida Property Tax Liability with Homestead Exemption

- Home value: H
 - Assessed Value: A
 - Millage rate: M
 - Tax = $A * M$
 - Where $A = f(H)$
- Three Types of Savings:
 1. Reduce A , assessed value with *Save our Homes* protection
 2. 50K of A exempt from school Millage
 3. 25K of A exemption from non-school Millage



Property Tax Liability with the Homestead Exemption



FL Statutes Section 196.061(1).

Property Tax Liability with the Homestead Exemption

Diagram illustrating the calculation of Total property tax liability (λ) with the Homestead Exemption, showing the impact of LRE (Local Revenue Exemption) and the "Save Our Home" (SOH) protections.

The formula for Total property tax liability (λ) is:

$$\lambda = (h - SOH - 50K) \cdot m_n + (h - SOH - 25K) \cdot m_s$$

Labels and arrows indicate the components and their relationships:

- Total property tax liability** points to λ .
- Home value** points to h .
- "Save Our Home" protections** points to SOH .
- Assessed Value** points to the terms $h - SOH$ in both parts of the formula.
- Non-School Millage** points to m_n .
- School Millage** points to m_s .
- LRE impact** is indicated by a bracket on the left, pointing to the SOH term.
- Reduce Taxable Value, transferable to another homestead** (1) points to the SOH term.
- \$50,000 exempt from non-school mills** (2) points to the $50K$ term.
- \$25,000 exempt from school mills** (3) points to the $25K$ term.

[SOH Detail >>](#)

FL Statutes Section 196.061(1).



**Darla Moore
School of Business**

Property Tax Liability with the Homestead Exemption

- Estimated Savings: \$2,000 annually on \$400K house*
 - ~\$167 per month
- *Save Our Homes* protections
 - Losing the exemption **resets assessed value to market value**

* House with \$100,000 of accrued SOH protections
FL Statutes Section 196.061(1).



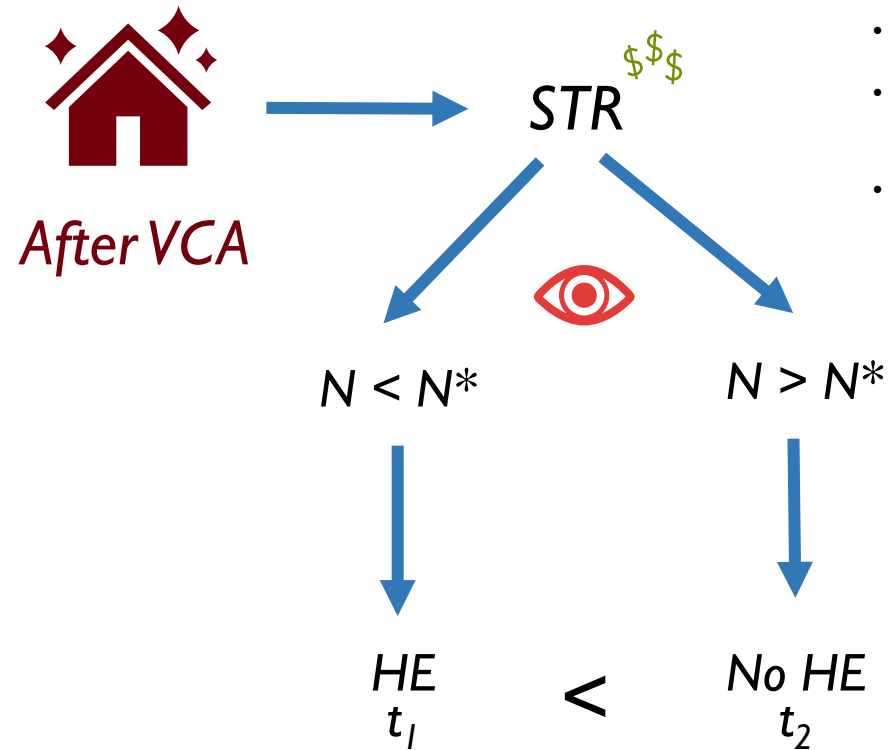
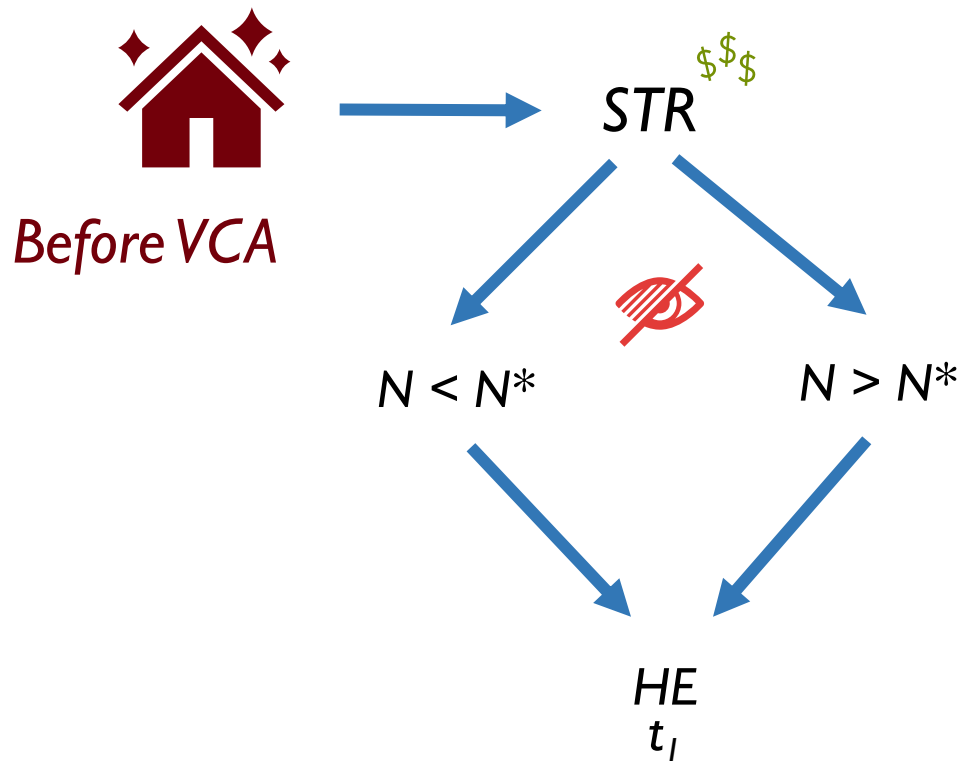
Darla Moore
School of Business

Airbnb Voluntary Collection Agreements

- Airbnb collects and remits tax to the local jurisdiction
- FL and Airbnb VCA: December 15, 2015



STR Reporting's Potential Impacts on Tax Exemption



- STR: short term rental (<30 days)
- N^* : annual rental day LRE limit
- LRE: Legal Residence Exemption – lowers property tax bill
- VCA: Voluntary Collection Agreement between Airbnb and the state

Motivation

Background

Empirical Strategy

Data

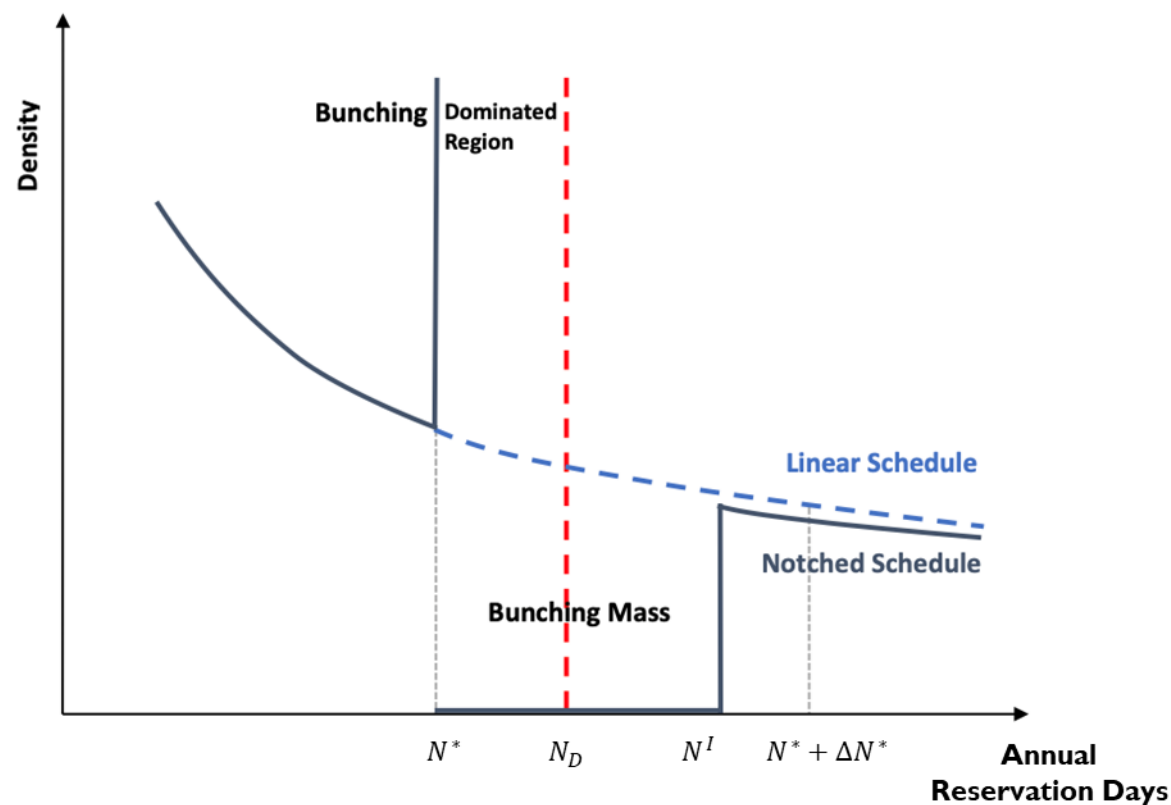
Results

Conclusion



**Darla Moore
School of Business**

A context for bunching estimation: the notch



- Model situations with incentive changes
- Isolate tax avoidance due to intentional behavior of STR hosts
- Detecting behavioral adjustments by clustering

Image from R Bunching package, [LINK](#)



Darla Moore
School of Business

Bunching estimation using notches in LRE rental limit

- Plot distribution of STR reservation days (N):
 - j bins with the count of STRs c_j in each bin
 - Bin: range of reservation days

Bunching estimation using notches in LRE rental limit

- Plot distribution of STR reservation days (N):
 - j bins with the count of STRs c_j in each bin
 - Bin: range of reservation days

$$c_j = \sum_{i=0}^p \beta_i (N_j)^i + \sum_{i=N_L}^{N_U} \gamma_i \mathbb{1}[N_j = i] + \sum_{r \in R} \rho_r \mathbb{1}\left[\frac{N_j}{r} \in \mathbb{N}\right] + \sum_{k \in K} \theta_k \mathbb{1}[N_j \in K \wedge N_j \notin [N_L, N_U]] + \epsilon_j$$

Diagram illustrating the components of the bunching estimation model:

- Binned data**: Points to the observed count c_j .
- Coefficients for polynomial**: Points to the polynomial term $\sum_{i=0}^p \beta_i (N_j)^i$.
- Bunching Region $[N_L, N_U]$** : Points to the discontinuity term $\sum_{i=N_L}^{N_U} \gamma_i \mathbb{1}[N_j = i]$.
- Round Number Controls (ex: 7 days)**: Points to the term $\sum_{r \in R} \rho_r \mathbb{1}\left[\frac{N_j}{r} \in \mathbb{N}\right]$.
- Fixed Number Control (ex: 14 days)**: Points to the term $\sum_{k \in K} \theta_k \mathbb{1}[N_j \in K \wedge N_j \notin [N_L, N_U]]$.
- Polynomial of degree p**: Points to the polynomial term.
- Allow discontinuity at $N = 30$** : Points to the bunching region.

Bunching estimation using notches in LRE rental limit

- Estimate Excess Mass at $N = 30$

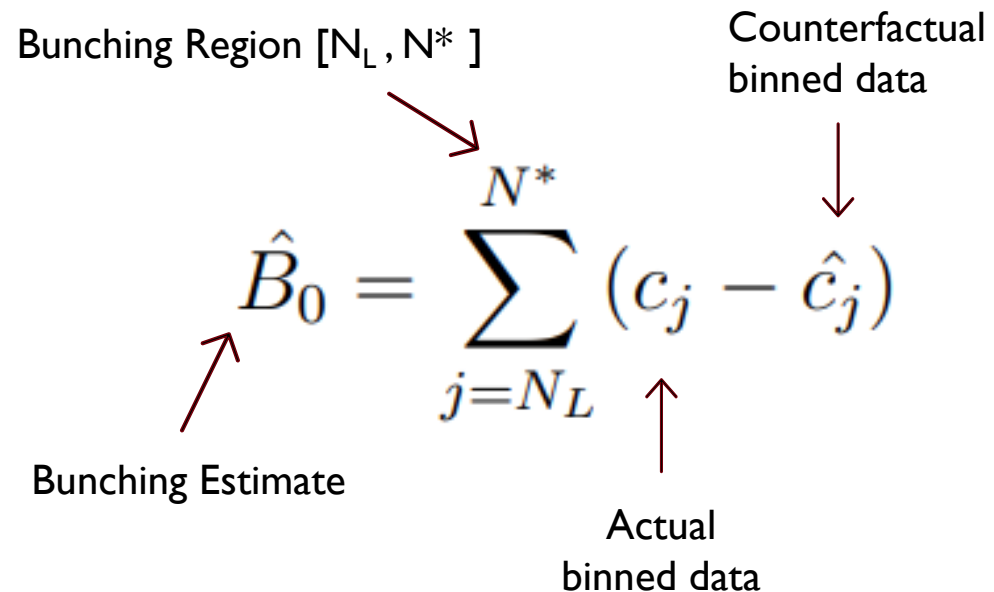
Bunching Region $[N_L, N^*]$

Counterfactual binned data

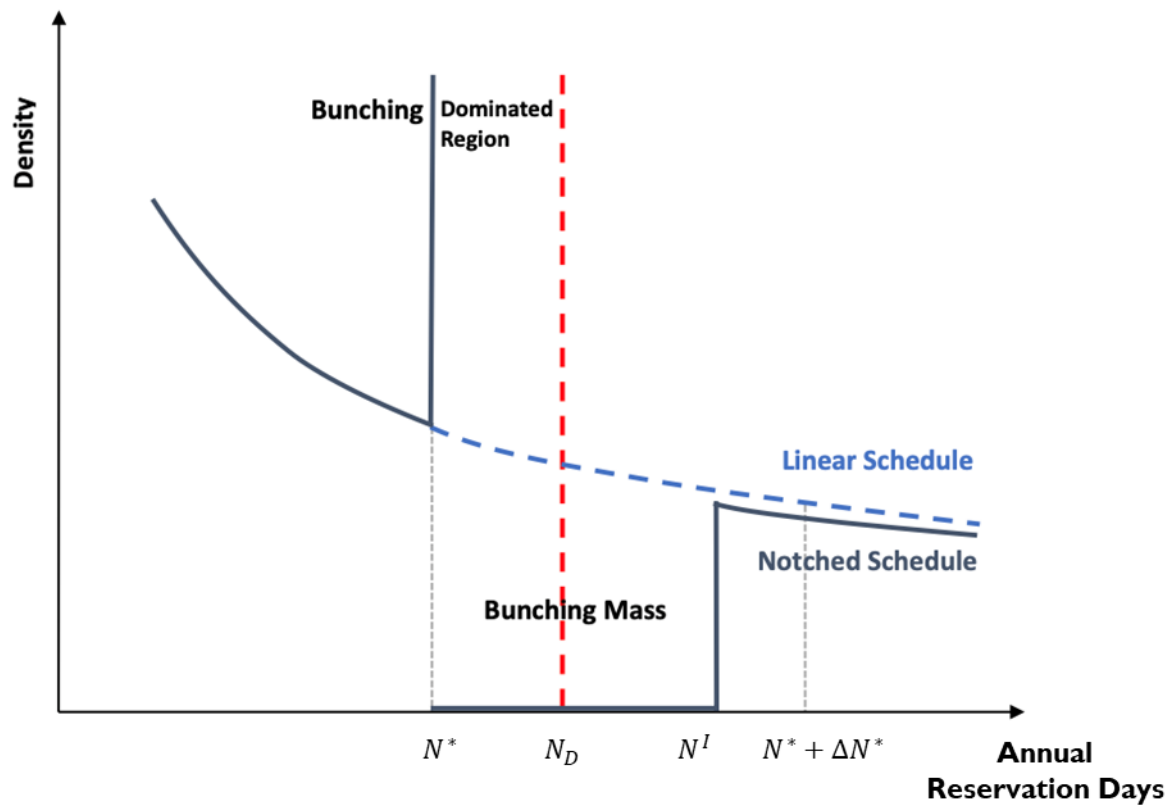
$$\hat{B}_0 = \sum_{j=N_L}^{N^*} (c_j - \hat{c}_j)$$

Bunching Estimate

Actual binned data



Bunching estimation using notches in LRE rental limit



- Normalized excess mass: reduction in nights

$$\hat{b} = \frac{\hat{B}_0}{\hat{c}_0}$$

- Local Average Treatment Effect (LATE)

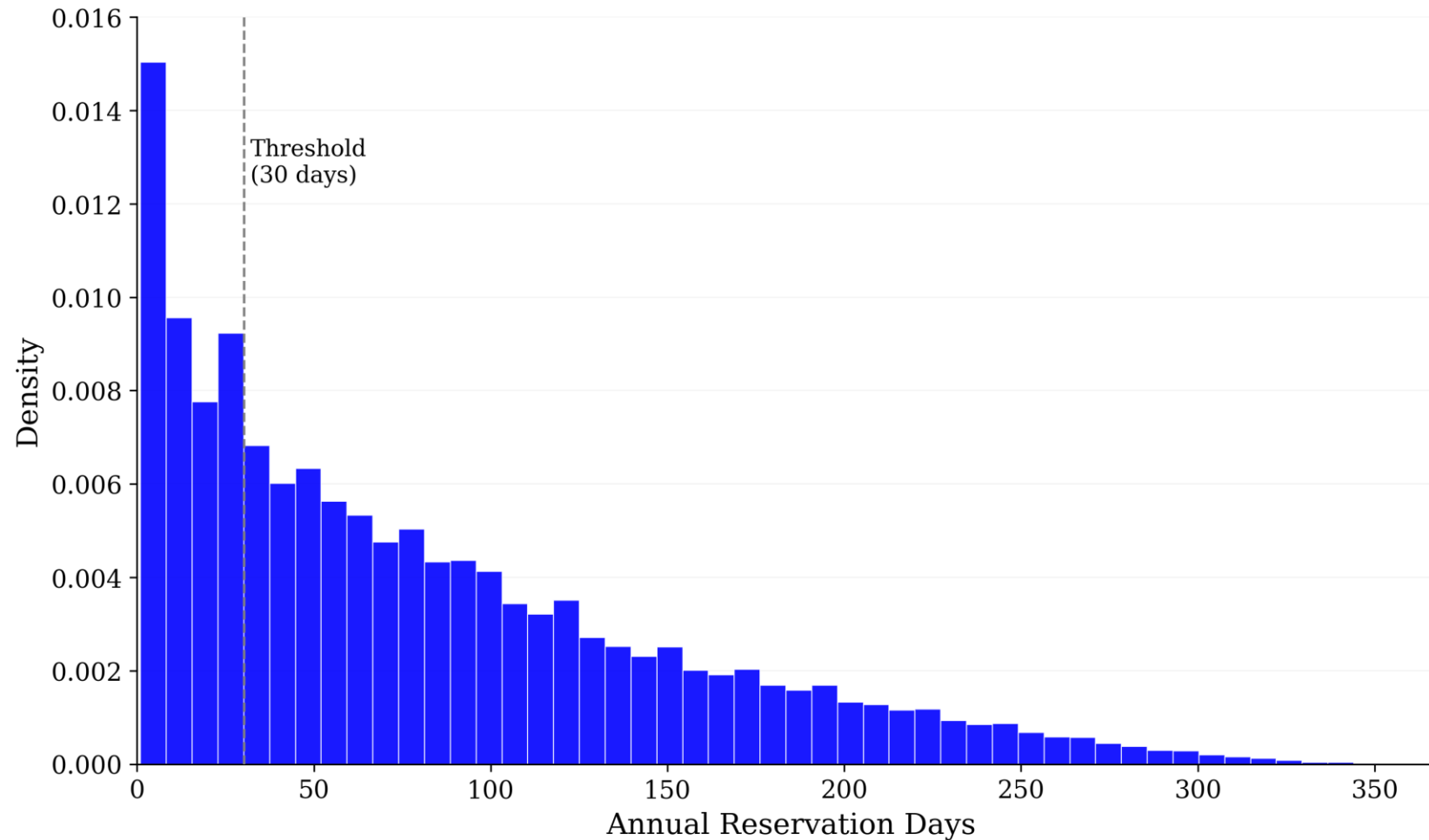
$$\hat{b}_{adj} = \frac{\hat{b}}{\widehat{sharecompliers}}$$

Bunching Estimator Assumptions*

1. Smooth distribution → changes are behavior responses
2. Bunchers come from above the threshold
3. The market has constant friction
4. No other changes at threshold
5. Correct functional form for counterfactual

* Kleven, H. J. and Waseem, M. (2013). Using notches to uncover optimization frictions and structural elasticities: Theory and evidence from Pakistan. *The Quarterly Journal of Economics*, 128(2):669–723.

Assumption 1: Smooth Distributions in STR market



Motivation

Background

Empirical Strategy

Data

Results

Conclusion



**Darla Moore
School of Business**

Data Sources

- AirDNA: 3rd party source for vacation rental data
 - Scraped Data and Partner Data
 - Monthly data from 2015 – 2024
- Voluntary Collection Agreements (VCA) – Bibler et al. (2021)*
 - Collect dates of VCAs
 - FL: December 15, 2015
- State Property Tax Law and Rates
 - FL Statutes section 196.061(1)
 - FL League of Cities



Sample Restrictions: Hosts likely to have LRE

- Entire Home Rentals
- Property Type
 - Likely Homeowner: Condo, House, Townhouse, Villa
 - Unlikely: Tent, Tipi, Yurt, Treehouse, Apartment
- Host Type
 - Host manages 1 Rental Unit



Motivation

Background

Empirical Strategy

Data

Results

Conclusion

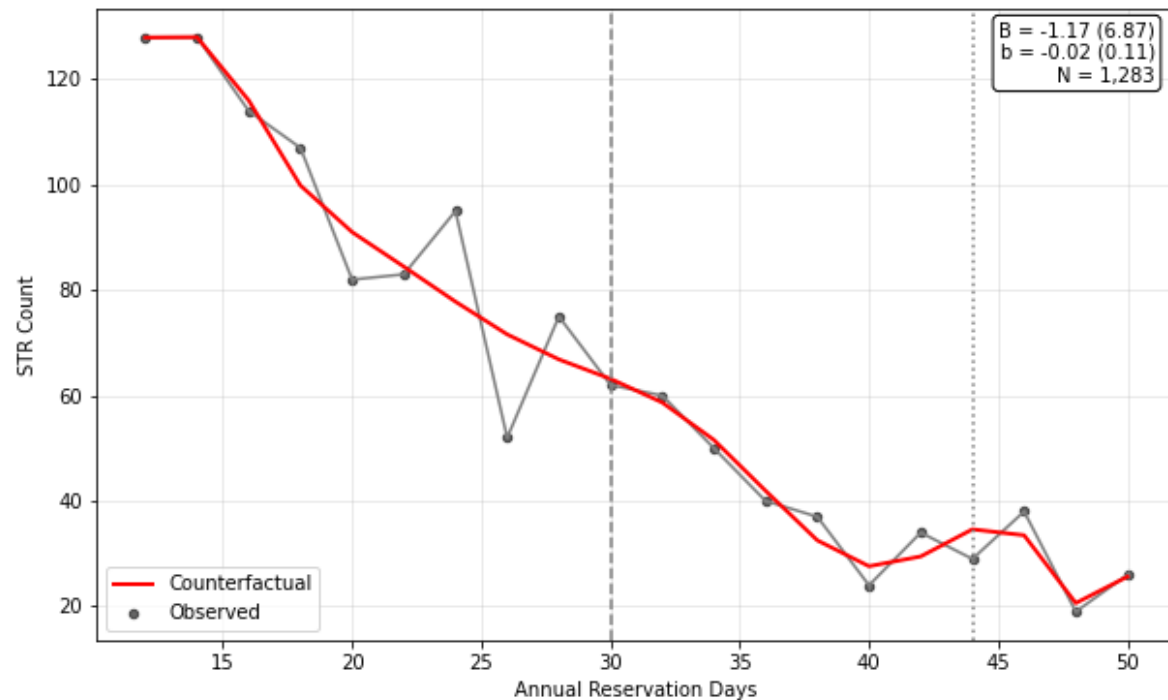


**Darla Moore
School of Business**

FL: Bunching Results pre & post-VCA

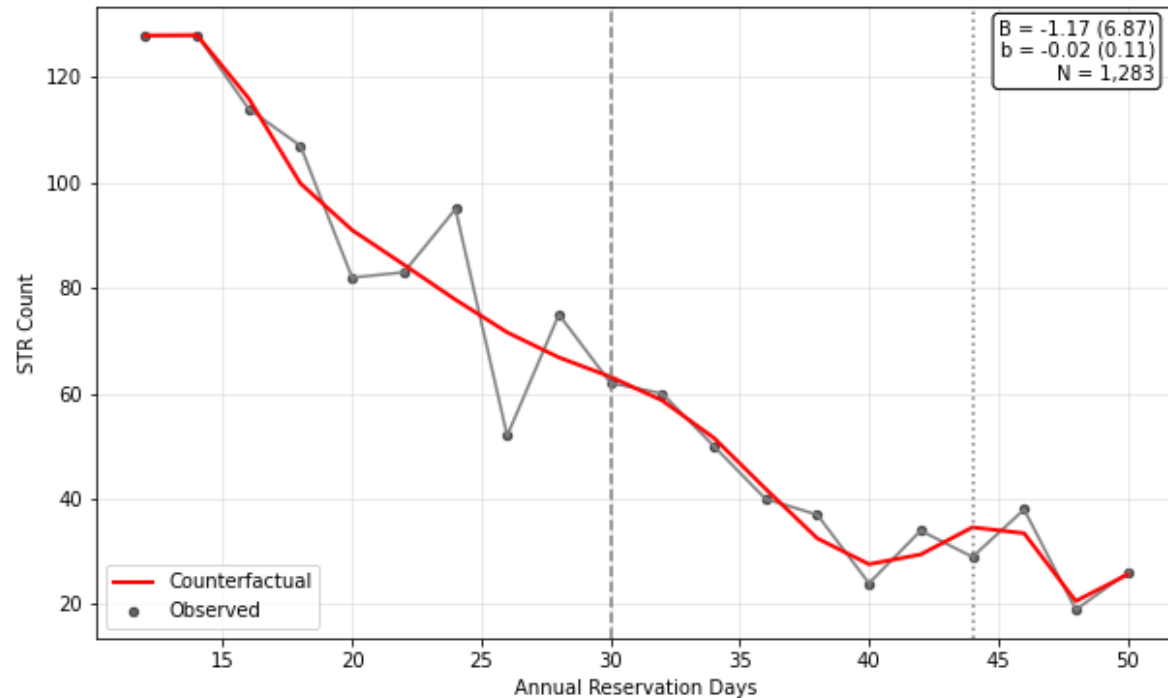
Pre-Period: 2015

Post-Period: 2017-2024

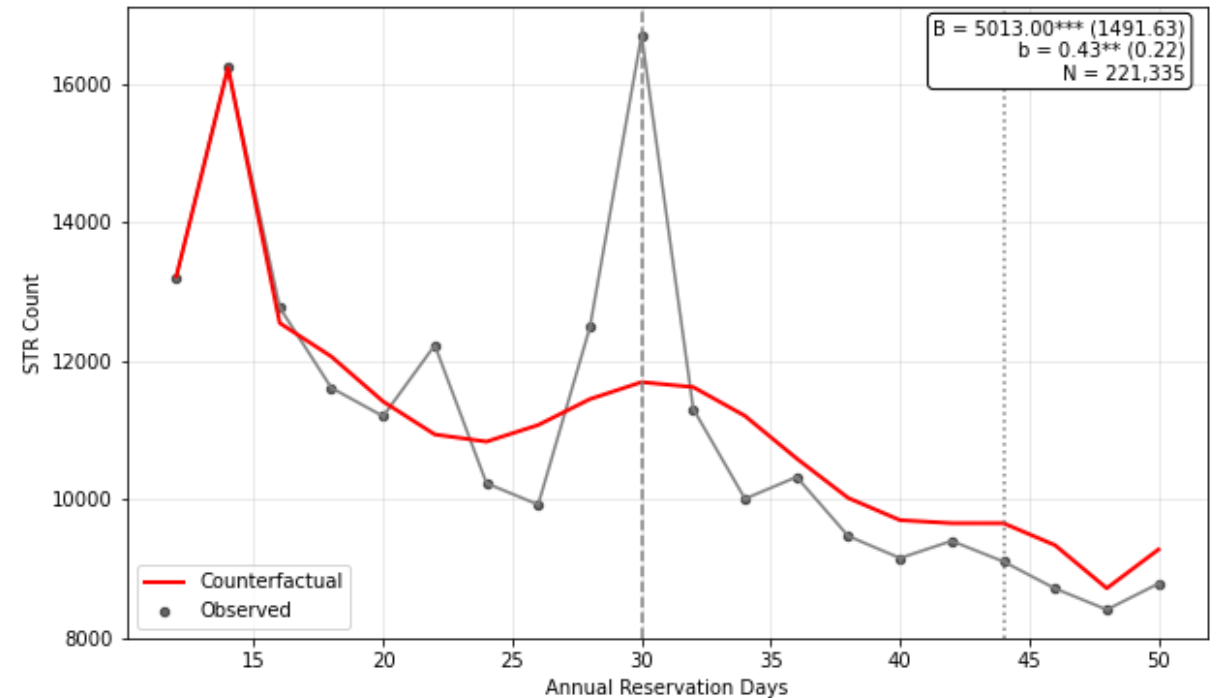


FL: Bunching Results pre & post-VCA

Pre-Period: 2015



Post-Period: 2017-2024



Bunching Results pre & post-VCA

State	Florida (30-day limit)	
Year(s)	2015	2017–2024
Estimated Excess Mass	-1.171 (7.957)	5,013.003*** (1,551.854)
Normalized Excess Mass	-0.019 (0.126)	0.429** (0.172)
Share of Bunchers	-0.022	0.027
Adj. Normalized Excess Mass (LATE)	0.826	15.793
N	1,283	221,335

Table 1: Bunching Estimation Results: Likely Homeowners with 1 Unit

Notes: This table reports bunching estimator results for the “Likely Homeowners with 1 Unit” subsample—STRs that are a townhouse, condo, house, or villa managed by a host with 1 unit. Data are from AirDNA. The columns estimate bunching at the 30-day rental limit (bin width: 2 days; 10 bins on each side; round-number control for 14). N is the number of observations in the bin range. Estimated excess mass is the number of additional STRs located at the threshold. Normalized excess mass is the average number of rental nights forgone at the cutoff. Share of bunchers is the fraction of STRs in the bunching region that respond to the cutoff (compliers). The adjusted normalized excess mass is the LATE—normalized excess mass divided by the share of bunchers—capturing how many rental nights the average buncher forgoes due to the cutoff. $p < 0.10^*$, $p < 0.05^{**}$, $p < 0.01^{***}$.

Quantifying the Response of the Marginal Buncher

- Elasticity = $\epsilon_{N,t} = \frac{\% \Delta N}{\% \Delta T} = \frac{\hat{b}/N^*}{\Delta \lambda / \lambda}$
 - 1% increase in property tax → 1.0-2.6% decrease in nights booked
 - **Compliers:** 1% increase in property tax → 36-101% decrease in nights booked

A 100% elasticity in Florida would be renting 29 days instead of 60 days.

Results Recap: How Do STR Hosts respond to property taxes?

- Bunching Response
 - FL Hosts reduce bookings by ~16 nights
- Post-Enforcement Change (VCA Effect)
 - Bunching becomes sharper after 2015 VCA, suggesting tighter compliance



Policy and Fiscal Implications

- STRs add service burdens but often erode tax bases
- Recoup costs through **enforceable** exemptions

Policy and Fiscal Implications

- STRs add service burdens but often erode tax bases
- Recoup costs through **enforceable** exemptions
- **Are counties equipped to enforce rental day limits?**
 - Counties face high enforcement costs
 - Data-sharing lowers enforcement costs
 - Integrate property tax records with STR data?

Policy and Fiscal Implications

- STRs add service burdens but often erode tax bases
- Recoup costs through **enforceable** exemptions
- **Are counties equipped to enforce rental day limits?**
 - Counties face high enforcement costs
 - Data-sharing lowers enforcement costs
 - Integrate property tax records with STR data?
- **Is 30 days the right limit? 14? 72?**
 - Tax design, salience, size impact compliance
 - Compliance → key to fairness and fiscal stability



THANK YOU

Kara Jones

Economics PhD Candidate

Kara.Jones@grad.moore.sc.edu

KJonesEcon.com

University of South Carolina



**Darla Moore
School of Business**

APPENDIX



**Darla Moore
School of Business**

STR Host and Taxpayer Optimization

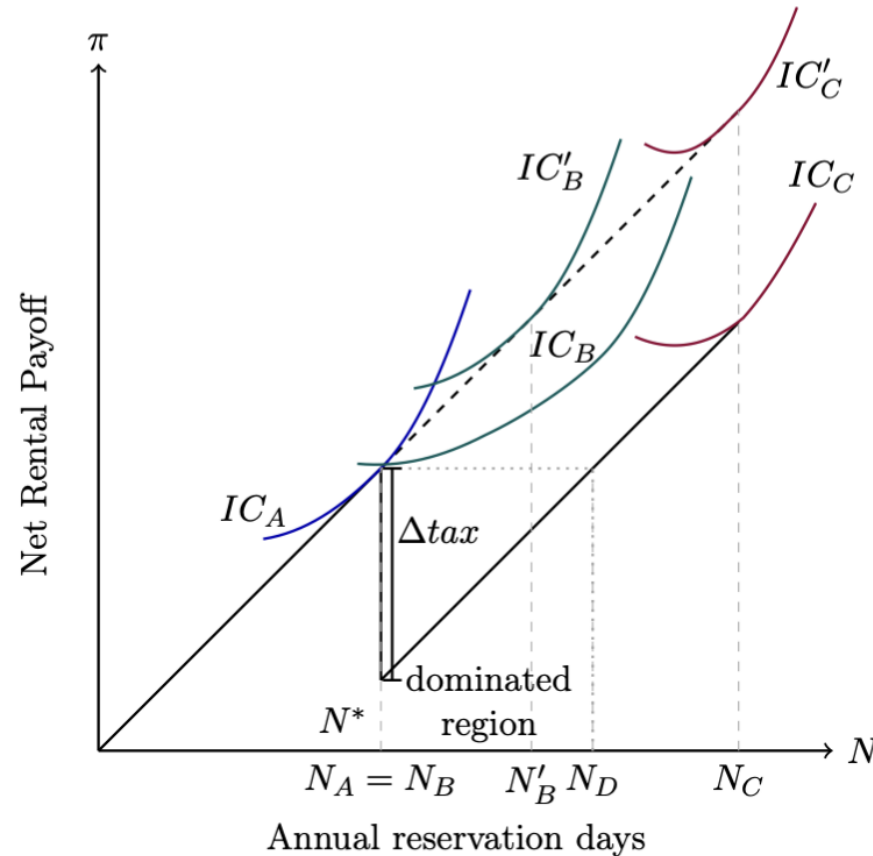
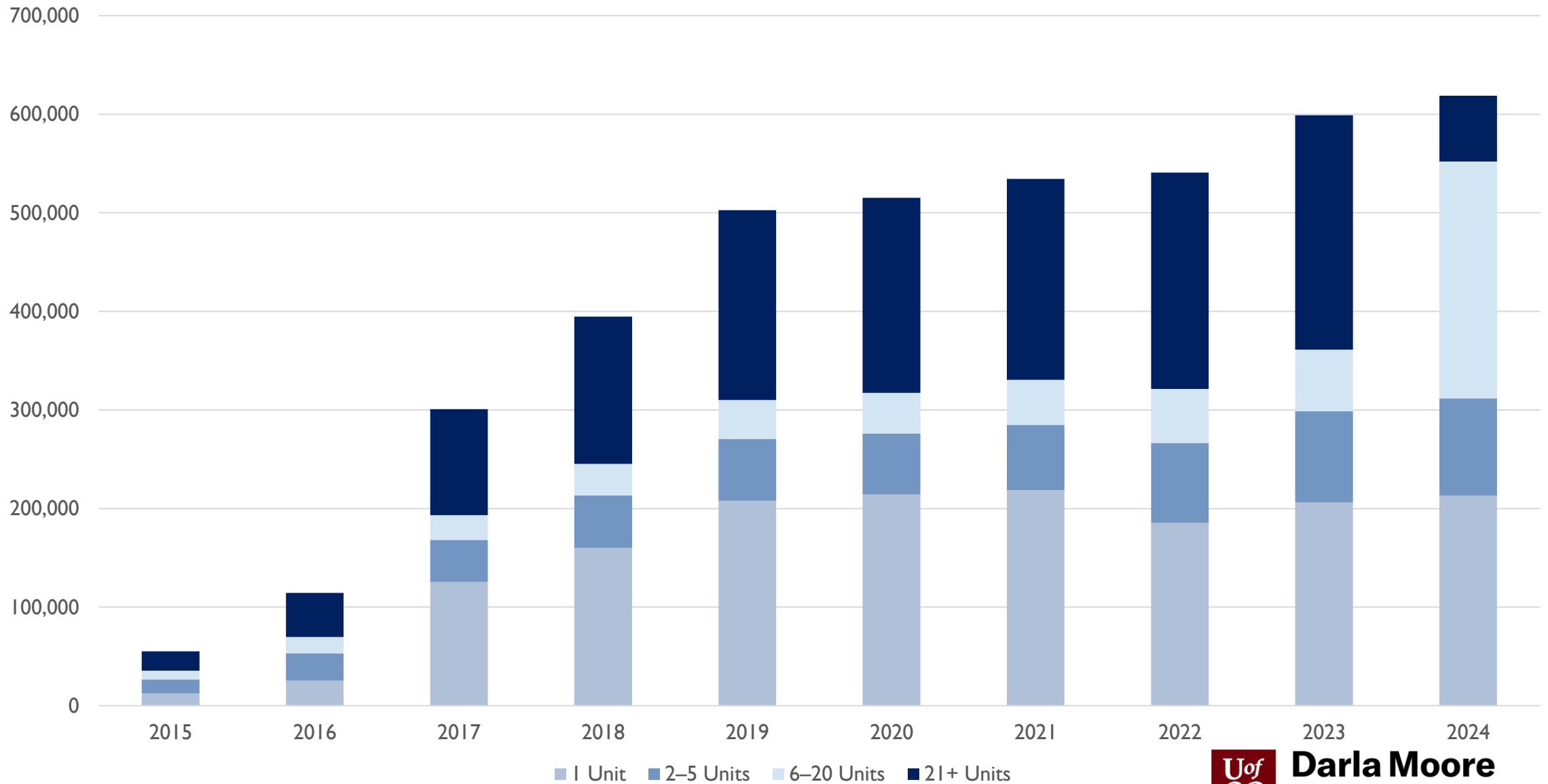


Figure 1: Bunching Incentives at Annual Rental Day Threshold

Growth in STRs in FL, from AirDNA data



Darla Moore
School of Business

Summary Statistics for 1-unit Hosts, 2015-2024

Variable	All of FL
<i>A. Property Characteristics</i>	
Num. of Bedrooms	3.07
Bathrooms	2.53
Share of House Listings	0.51
Share of Condo Listings	0.33
Share of Properties on Airbnb and Vrbo	0.09
Years as a STR	4.78
<i>B. Other Characteristics</i>	
Annual Number of Reservations	30.43
Annual Reservation Days	142.46
Average Occupancy Rate	0.53
Annual Revenue in Dollars	31,096.68
Revenue / Revenue Potential	0.81
Average Nightly Rate	220.46
Observations	1,311,273



**Darla Moore
School of Business**

Other states with exemptions and rental limits

Table 1: Property Tax Overview by State

State	Principal Residence Exemption Amount	Rental Day Limit per year for PRHE	Year of State VCA
Florida	\$50,000 reduction in assessed value.	30	2015
Michigan	Up to 18 mills reduction in millage rate.	14	2017
South Carolina	Decrease 6% assesement ratio to 4%.	72	2016
Vermont	Decrease tax rate of 1.83% to 1%.	182	2016

